

205/210 QUICK START GUIDE

Standard Operating Procedure for AAS Machine

- Turn on gas cylinders.
- Turn on instrument (Rocker switch in back and Red switch on right side of 210).
- Ensure master tube is connected with a water filled trap in it.
- Place ignition button (IBC) in one of the lamp openings on the left hand side of the 210.
- Type element.
- Press “Set” until the lamp # on the top of the screen matches the lamp position of the HCL.
- Scroll through Libraries until the element of interest is found (lamp wavelength and possible “D2” for flame analysis in library name, “hyd” and “hyd” for hydride generation analysis).
- Press “Esc” to load library, then “Enter”.
- Press “Align”.
- Set slit-width dial to recommended value from the top of the screen.
- Set monochromator dial to desired wavelength (recommended wavelength is listed on top of 210 screen).
- Slowly rotate monochromator dial until the sample energy bar on the screen is as far right as possible. If the bar goes off the screen, press “Align” to renormalize the bar graph and continue optimizing.
- Adjust the vertical and horizontal knobs on the lamp turret to get maximum energy on the alignment screen. Usually sample energy is about 35 for flame analysis or 0.2 for furnace analysis.
- Press “AZ”.
- If using front knob to “air”.
- Ignite flame with lighter.
- If using Nitrous flame, increase fuel knob so that a bright yellow flame is produced and turn knob from “air” to “N2O”.
- Aspirate a standard of the metal to be analyzed and adjust the burner vertical and horizontal for maximum absorbance. (If the absorbance goes above 1.2, use a lower concentration standard).

- With the still aspirating standard, adjust the fuel knob until maximum absorbance is achieved. (You can also adjust the nebulizer to maximize absorbance, but this does not need to be done every time).
- Aspirate blank and press “A/Z” when the absorbance is stable.
- The instrument is ready for analysis, refer to manual or below for calibrating the instrument.

If using Nitrous Oxide/acetylene, turn knob from “N₂O” to “Air”, shut off fuel switch, turn knob to “off”.

If using air/acetylene, shut off fuel switch, turn knob from “Air” to “off”.

CALIBRATION PROCEDURE FOR AAS MACHINE

- Press “Cal” button to bring up calibration screen.
- Press “Cls” button to bring up calibration controls screen. Change units to desired value (ppm most common). Change curve fit to appropriate order. Linear requires at least 1 standard for calibration, 2nd order requires at least 2 standards for calibration, but at least one more is recommended to check for proper fit.
- Press “Esc”.
- Press “Sel” to highlight the second row (first standard, not blank).
- Press “Sel” to edit concentration value.
- Press “<” or “>” for the concentration of the standard.
- Press “↓” to highlight the next row and enter the concentration value for the next standard (the order of standards does not matter).
- Repeat as many times as necessary for all of your standards.
- If more rows for standards are needed, press “Sel” until cup number is highlighted, go to the first empty row and enter the next cup number in sequence, press “Enter”. Repeat as needed.
- If too many rows are present for the number of standards required, go to the unnecessary cup number, delete the number, and press “Enter”.
- Press “Start”.
- Follow instructions on the screen for when to aspirate the standards and when to press “Enter”. When all data has been entered, press “Cal” again, review the screen and press “Yes” to calculate the calibration curve.
- Press “Esc” twice to plot the calibration curve. Ensure that it is acceptable.
- Press “Esc” twice to return to the main screen. The instrument is now calibrated and ready to read out directly in concentration for samples.